Multiple Linear Regression

1 From SLR to MLR

Recap: simple linear regression

The model with one regressor:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n.$$

OLS estimators of the coefficients, with $S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y})$ and $S_{xx} = \sum_i (x_i - \bar{x})^2$, and the unbiased estimator of the error variance:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \tag{1}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

$$\hat{\sigma}^2 = \frac{\mathrm{SS}_{\mathrm{Res}}}{n-2}. \tag{2}$$

Question: is one regressor enough?

Delivery time of a soft-drink route driver depends on the number of cases loaded (x_1) and the walking distance (x_2).

Why not simply fit two separate SLRs, y on x_1 and y on x_2?

- The two regressors are typically correlated (more cases usually means longer distance).
- Each SLR slope then absorbs part of the effect of the omitted regressor, so it is not the effect of that regressor alone.
- Separate fits cannot answer "what is the effect of one more case for a fixed distance?".
- The residual variance is also larger because the omitted regressor's effect is left in the error.

Example: delivery time data

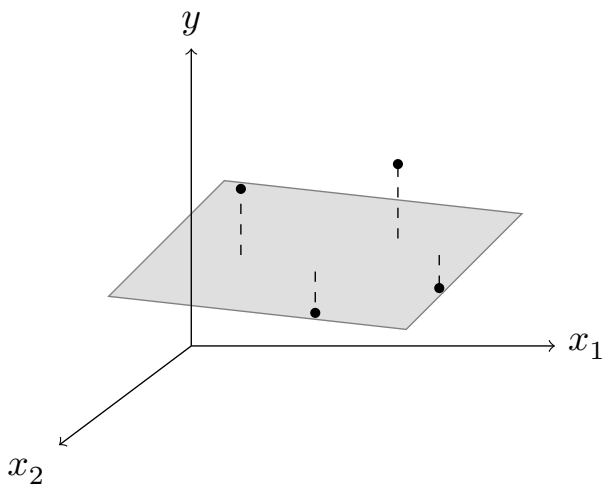
Montgomery, Peck & Vining, Example 3.1 ($n = 25$).

- y: delivery time (minutes)
- x_1: number of cases stocked
- x_2: distance walked by the driver (feet)

Obs.	Time y	Cases x_1	Distance x_2
1	16.68	7	560
2	11.50	3	220
3	12.03	3	340
⋮	⋮	⋮	⋮
25	10.75	4	150

From a line to a plane

- SLR: a line in the (x, y) plane.
- Two regressors: a plane in (x_1, x_2, y) space.
- k regressors: a hyperplane in $(k + 1)$ dimensions.
- Residuals: vertical distances to the fitted surface.



2 The MLR model

The MLR equation

With k regressors and n observations:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i, \quad i = 1, \ldots, n. \tag{3}$$

- x_{ij}: value of regressor j for observation i.
- $p = k + 1$ unknown regression coefficients $\beta_0, \ldots, \beta_k$ (including the intercept).
- ε_i: random error of observation i.

Meaning of the coefficients

$$\frac{\partial \, \mathrm{E}[y \mid x_1, \ldots, x_k]}{\partial x_j} = \beta_j, \quad j = 1, \ldots, k.$$

- β_j is the change in the mean response per unit increase in x_j *when all other regressors are held fixed.*
- For this reason $\beta_1, \ldots, \beta_k$ are called partial regression coefficients.
- β_0 is the mean response when all regressors are zero (meaningful only if that point is within the range of the data).

Question: interpreting the coefficients

In the delivery time model $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$, interpret β_1, β_2 and β_0 in words, with units.

From n equations to one matrix equation

Writing (3) out for every observation:

$$\begin{aligned} y_1 &= \beta_0 + \beta_1 x_{11} + \cdots + \beta_k x_{1k} + \varepsilon_1 \\ y_2 &= \beta_0 + \beta_1 x_{21} + \cdots + \beta_k x_{2k} + \varepsilon_2 \\ &\vdots \\ y_n &= \beta_0 + \beta_1 x_{n1} + \cdots + \beta_k x_{nk} + \varepsilon_n \end{aligned}$$

Stack the responses, the coefficients and the errors into vectors, and the regressor values into a matrix.

The MLR model in matrix form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \tag{4}$$

$$\underset{n\times 1}{\mathbf{y}}, \qquad \underset{n\times p}{\mathbf{X}}, \qquad \underset{p\times 1}{\boldsymbol{\beta}}, \qquad \underset{n\times 1}{\boldsymbol{\varepsilon}}.$$

The vectors y, $\boldsymbol{\beta}$, $\boldsymbol{\varepsilon}$

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}, \qquad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

The design matrix X

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix}$$

- Row i is observation i; column j is regressor j.
- The first column of ones corresponds to the intercept β_0.

Model assumptions

$$\mathrm{E}[\boldsymbol{\varepsilon}] = \mathbf{0}, \tag{5}$$

$$\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n. \tag{6}$$

- $\mathbf{X}$ is fixed (non-random) and has full column rank, i.e. $\mathrm{rank}(\mathbf{X}) = p \leq n$.
- No distributional assumption on $\boldsymbol{\varepsilon}$ is needed yet for estimation (however noramility assumption is required later for inference)

$\mathrm{E}[\boldsymbol{\varepsilon}]$ is a n dimensional vector of zeroes.

What does $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$ say?

$$\mathrm{Var}(\boldsymbol{\varepsilon}) = \mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'] = \mathrm{E}[(\boldsymbol{\varepsilon} - \mathbf{0})(\boldsymbol{\varepsilon} - \mathbf{0})'] = \begin{pmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{pmatrix}$$

- $\mathrm{Var}(\boldsymbol{\varepsilon})$ is $n \times n$ dimensional variance-covariance matrix.
- Diagonal entries: the i-th diagonal entry contains the variance of the i-th error term. In this model, every error has the same variance σ^2 (constant variance).
- Off-diagonal entries: cell (i, l) contains the covariance between observations i and l. In this model, errors of different observations are uncorrelated, hence covariances are zero.

"Linear" means linear in the parameters

Each of these is an MLR model:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon,$$
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \varepsilon.$$

This one is not:

$$y = \beta_0 e^{\beta_1 x} + \varepsilon.$$

Polynomial and interaction terms are handled later in the course.

3 OLS estimation in matrix form

The least-squares criterion

Choose $\boldsymbol{\beta}$ to minimise the sum of squared errors

$$S(\boldsymbol{\beta}) = \sum_{i=1}^{n} \varepsilon_i^2 = \boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}). \tag{7}$$

Expanding $S(\boldsymbol{\beta})$

$$\begin{aligned} S(\boldsymbol{\beta}) &= \mathbf{y}'\mathbf{y} - \boldsymbol{\beta}'\mathbf{X}'\mathbf{y} - \mathbf{y}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} \\ &= \mathbf{y}'\mathbf{y} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}. \end{aligned} \tag{8}$$

The second line combines the two middle terms.

Why is $\mathbf{y}'\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}'\mathbf{X}'\mathbf{y}$?

Two rules for differentiating with respect to a vector

For a constant vector $\mathbf{a}$ and a constant symmetric matrix $\mathbf{A}$:

$$\frac{\partial\, \mathbf{a}'\boldsymbol{\beta}}{\partial \boldsymbol{\beta}} = \mathbf{a}, \qquad \frac{\partial\, \boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta}}{\partial \boldsymbol{\beta}} = 2\mathbf{A}\boldsymbol{\beta}.$$

Apply with $\mathbf{a} = \mathbf{X}'\mathbf{y}$ and $\mathbf{A} = \mathbf{X}'\mathbf{X}$ (symmetric) to (8).

The normal equations

$$\begin{aligned}
\frac{\partial S}{\partial \boldsymbol{\beta}} &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\boldsymbol{\beta}, \\
\frac{\partial S}{\partial \boldsymbol{\beta}}\Big|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}} &= -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0}, \\
\Rightarrow \mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} &= \mathbf{X}'\mathbf{y}. \qquad (9)
\end{aligned}$$

This is a system of p linear equations in the p unknowns $\hat{\beta}_0, \ldots, \hat{\beta}_k$.

The OLS estimator

If $\mathbf{X}'\mathbf{X}$ is invertible, the normal equations (9) have the unique solution

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}. \qquad (10)$$

When is $\mathbf{X}'\mathbf{X}$ invertible?

- $\mathbf{X}'\mathbf{X}$ $(p \times p)$ is invertible if and only if $\mathbf{X}$ has full column rank.
- That requires $n \geq p$ and no column of $\mathbf{X}$ being a linear combination of the other columns.
- If a regressor is (nearly) a linear combination of the others, $\mathbf{X}'\mathbf{X}$ is singular (nearly singular): this is the collinearity problem, discussed in a later lecture.

Is $\hat{\boldsymbol{\beta}}$ a minimum?

$$\frac{\partial^2 S}{\partial \boldsymbol{\beta}\, \partial \boldsymbol{\beta}'} = 2\mathbf{X}'\mathbf{X}.$$

- For any $\mathbf{a} \neq \mathbf{0}$: $\mathbf{a}'\mathbf{X}'\mathbf{X}\mathbf{a} = (\mathbf{X}\mathbf{a})'(\mathbf{X}\mathbf{a}) = \|\mathbf{X}\mathbf{a}\|^2 > 0$, because full column rank gives $\mathbf{X}\mathbf{a} \neq \mathbf{0}$.
- So $\mathbf{X}'\mathbf{X}$ is positive definite and $\hat{\boldsymbol{\beta}}$ is the unique minimiser of (7).

4 Fitted values, residuals and the hat matrix

Fitted values and residuals

$$\begin{aligned}
\hat{\boldsymbol{\beta}} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\
\hat{\mathbf{y}} &= \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{y}, \\
\mathbf{H} &= \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \qquad (11) \\
\mathbf{e} &= \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}. \qquad (12)
\end{aligned}$$

$\mathbf{H}$ is the $n \times n$ hat matrix: it puts the hat on $\mathbf{y}$.

Properties of the hat matrix

$$\mathbf{H}' = \mathbf{H}, \tag{13}$$
$$\mathbf{H}^2 = \mathbf{H}, \tag{14}$$
$$\mathbf{HX} = \mathbf{X}. \tag{15}$$

That is, $\mathbf{H}$ is symmetric and idempotent.

Prove (14) and (15) from the definition (11).

Properties of $\mathbf{I} - \mathbf{H}$

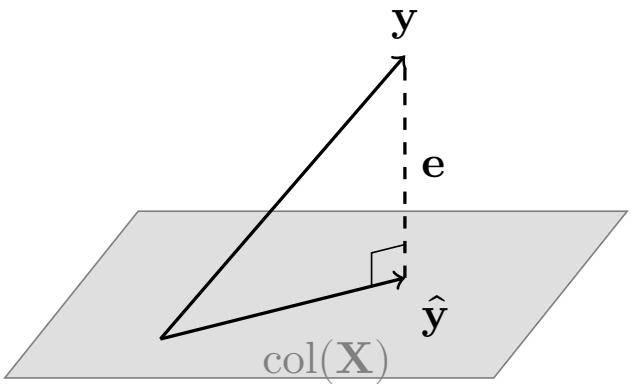
From (13), (14) and (15):

$$(\mathbf{I} - \mathbf{H})' = \mathbf{I} - \mathbf{H}, \tag{16}$$
$$(\mathbf{I} - \mathbf{H})^2 = \mathbf{I} - \mathbf{H}, \tag{17}$$
$$(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{0}. \tag{18}$$

Geometry of least squares

$\mathbf{H}$ is symmetric and idempotent and $\hat{\mathbf{y}} = \mathbf{Hy}$, therefore, $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto $\mathbf{X}$.

- The columns of $\mathbf{X}$ span a p-dimensional subspace $\text{col}(\mathbf{X})$ of $\mathbb{R}^n$.
- $\hat{\mathbf{y}} = \mathbf{Hy}$ is the orthogonal projection of $\mathbf{y}$ onto $\text{col}(\mathbf{X})$.
- $\mathbf{e}$ is perpendicular to $\text{col}(\mathbf{X})$.
- $\hat{\mathbf{y}}$ is the point of $\text{col}(\mathbf{X})$ closest to $\mathbf{y}$, so $\hat{\boldsymbol{\beta}}$ minimises $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$.



Orthogonality of residuals and regressors

$$\mathbf{X}'\mathbf{e} = \mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{X}'\mathbf{y} - \mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0}, \tag{19}$$

by the normal equations (9). Consequences:

- First row of $\mathbf{X}'$ is a row of ones, so $\sum_i e_i = 0$.
- Row j gives $\sum_i x_{ij} e_i = 0$ for each regressor.
- $\hat{\mathbf{y}}'\mathbf{e} = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{e} = 0$.

5 Delivery time example: OLS fit

Computing $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}'\mathbf{y}$

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} 25 & 219 & 10\,232 \\ 219 & 3\,055 & 133\,899 \\ 10\,232 & 133\,899 & 6\,725\,688 \end{pmatrix},$$

$$\mathbf{X}'\mathbf{y} = \begin{pmatrix} 559.60 \\ 7\,375.44 \\ 337\,071.69 \end{pmatrix}.$$

The inverse

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{pmatrix} 1.13 \times 10^{-1} & -4.45 \times 10^{-3} & -8.37 \times 10^{-5} \\ -4.45 \times 10^{-3} & 2.74 \times 10^{-3} & -4.79 \times 10^{-5} \\ -8.37 \times 10^{-5} & -4.79 \times 10^{-5} & 1.23 \times 10^{-6} \end{pmatrix}$$

The fitted equation

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \begin{pmatrix} 2.3412 \\ 1.6159 \\ 0.01438 \end{pmatrix}$$

$$\hat{y} = 2.3412 + 1.6159\,x_1 + 0.01438\,x_2.$$

Interpreting the fit

- Each additional case adds about 1.62 minutes to the delivery time, for a fixed distance.
- Each additional foot adds about 0.0144 minutes (about 1.4 minutes per 100 ft), for a fixed number of cases.
- The SLR of y on x_1 alone gives a slope of 2.18, not 1.62: cases and distance are correlated ($r = 0.82$), and the SLR slope absorbs part of the distance effect.

Verifying in R

```
y <- delivery$time
X <- cbind(1, delivery$cases, delivery$distance)

solve(t(X) %*% X, t(X) %*% y)

coef(lm(time ~ cases + distance,
        data = delivery))
```

- `solve(A, b)` solves $\mathbf{A}\mathbf{x} = \mathbf{b}$ directly; it is numerically preferable to forming $\mathbf{A}^{-1}$ explicitly.

6 Properties of the OLS estimators

Linearity

$$\hat{\boldsymbol{\beta}} = \mathbf{A}\mathbf{y}, \qquad \mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'. \tag{20}$$

$\mathbf{A}$ is a $p \times n$ matrix of constants (given $\mathbf{X}$), so every $\hat{\beta}_j$ is a linear combination of $y_1, \dots, y_n$.

Decomposing $\hat{\boldsymbol{\beta}}$

Substitute the model (4) into (10):

$$\begin{aligned}
\hat{\boldsymbol{\beta}} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\
\hat{\boldsymbol{\beta}} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\
&= \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}.
\end{aligned} \tag{21}$$

The estimator is the true $\boldsymbol{\beta}$ plus a linear function of the errors.

Unbiasedness

Taking expectations in (21) with $\mathbf{X}$ fixed, and using (5):

$$\begin{aligned}
\mathrm{E}[\hat{\boldsymbol{\beta}}] &= \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\mathrm{E}[\boldsymbol{\varepsilon}] \\
&= \boldsymbol{\beta}.
\end{aligned}$$

The covariance matrix of a random vector

$\hat{\boldsymbol{\beta}}$ is p dimensional random vector. Therefore, $\mathrm{Var}(\hat{\boldsymbol{\beta}})$ is a $p \times p$ symmetric matrix:

- diagonal entries: $\mathrm{Var}(\hat{\beta}_j)$;
- off-diagonal entries: $\mathrm{Cov}(\hat{\beta}_i, \hat{\beta}_j)$.

For a constant matrix $\mathbf{B}$ and a random vector $\mathbf{z}$, directly from the definition,

$$\mathrm{Var}(\mathbf{B}\mathbf{z}) = \mathbf{B}\ \mathrm{Var}(\mathbf{z})\,\mathbf{B}'. \tag{22}$$

Covariance matrix of $\hat{\boldsymbol{\beta}}$

$$\begin{aligned}
\mathrm{Var}(\hat{\boldsymbol{\beta}}) &= \mathrm{Var}\left(\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}\right) \\
&= \mathrm{Var}\left((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}\right) \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\ \mathrm{Var}(\boldsymbol{\varepsilon})\,\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\
&= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\
&= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.
\end{aligned} \tag{23}$$

The steps use (21) (the constant $\boldsymbol{\beta}$ adds no variance), (22) and (6).

Reading the covariance matrix

Let $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}$, with entries C_{ij}, $i, j = 0, \dots, k$.

$$\begin{aligned} \mathrm{Var}(\hat{\beta}_j) &= \sigma^2 C_{jj}, \\ \mathrm{Cov}(\hat{\beta}_i, \hat{\beta}_j) &= \sigma^2 C_{ij}, \quad i \neq j. \end{aligned} \tag{24}$$

Question: what makes $\mathrm{Var}(\hat{\beta}_j)$ large?

Using (24), which features of the data and the design make the variance of $\hat{\beta}_j$ large?

- Large error variance σ^2.
- Small sample size n.
- Little spread in regressor $\mathbf{x}_j$ (as in SLR, $\mathrm{Var}(\hat{\beta}_1) = \sigma^2/S_{xx}$).
- Strong linear relationship between $\mathbf{x}_j$ and the other regressors: C_{jj} becomes large. This is the collinearity problem.

6.1 Distribution of the estimates under normal errors

Distribution of $\hat{\boldsymbol{\beta}}$

Add the assumption $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$. By (21), $\hat{\boldsymbol{\beta}}$ is a linear function of a normal vector, hence normal:

$$\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta},\ \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}). \tag{25}$$

The mean and covariance matrix are the ones derived above. This result drives the inference procedures in the coming weeks.

7 The Gauss–Markov theorem

Gauss–Markov theorem

Under (5) and (6) (no normality needed), the OLS estimator $\hat{\boldsymbol{\beta}}$ is the best linear unbiased estimator (BLUE) of $\boldsymbol{\beta}$.

- Linear: a linear function of $\mathbf{y}$.
- Unbiased: $\mathrm{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$.
- Best: among all linear unbiased estimators, each $\hat{\beta}_j$, and more generally each $\mathbf{c}'\hat{\boldsymbol{\beta}}$, has the smallest variance.

8 Maximum Likelihood Estimation of MLR Parameters

Maximum likelihood under normal errors

The likelihood function is given by:

$$\begin{aligned} L(\boldsymbol{\beta}, \sigma^2) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\!\left(-\frac{(y_i - \mathbf{x}_i'\boldsymbol{\beta})^2}{2\sigma^2} \right) \\ &= (2\pi\sigma^2)^{-n/2} \exp\!\left(-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \right). \end{aligned}$$

The log-likelihood of $(\boldsymbol{\beta}, \sigma^2)$ is

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}). \tag{26}$$

For any fixed σ^2, maximising over $\boldsymbol{\beta}$ is the same as minimising $S(\boldsymbol{\beta})$ in (7), so the ML estimator of $\boldsymbol{\beta}$ equals the OLS estimator.

Maximum likelihood estimator of σ^2

Setting $\partial\ell/\partial\sigma^2 = 0$ at $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$ gives

$$\hat{\sigma}^2_{\mathrm{ML}} = \frac{(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})}{n} = \frac{\mathrm{SS}_{\mathrm{Res}}}{n}. \tag{27}$$

We show next that this estimator is biased.

9 Estimating the variance parameter σ^2

Residuals as a function of the errors

The residual vector can also be written as:

$$\begin{aligned}\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{H}\mathbf{y} &= (\mathbf{I} - \mathbf{H})\mathbf{y} \\ &= (\mathbf{I} - \mathbf{H})(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon},.\end{aligned} \tag{28}$$

Equation (28) holds because $(\mathbf{I} - \mathbf{H})\mathbf{X} = 0$ as per (18). Then the residual sum of squares:

$$\begin{aligned}\mathrm{SS}_{\mathrm{Res}} = \sum_{i=1}^{n} e_i^2 &= \mathbf{e}'\mathbf{e} \\ &= \boldsymbol{\varepsilon}'(\mathbf{I} - \mathbf{H})'(\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}'(\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}.\end{aligned} \tag{29}$$

Equation (29) holds because $(\mathbf{I} - \mathbf{H})$ is symmetric and idempotent as per equations (16) and (17).

Expectation of a quadratic form

For a constant symmetric matrix $\mathbf{B}$:

$$\begin{aligned}\mathrm{E}[\boldsymbol{\varepsilon}'\mathbf{B}\boldsymbol{\varepsilon}] &= \mathrm{E}\left[\,\mathrm{tr}(\boldsymbol{\varepsilon}'\mathbf{B}\boldsymbol{\varepsilon})\right] = \mathrm{E}\left[\,\mathrm{tr}(\mathbf{B}\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}')\right] \\ &= \mathrm{tr}\left(\mathbf{B}\ \mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}']\right) = \sigma^2\ \mathrm{tr}(\mathbf{B}).\end{aligned} \tag{30}$$

- A scalar equals its trace.
- $\mathrm{tr}(\mathbf{MN}) = \mathrm{tr}(\mathbf{NM})$, and expectation (E) commutes with tr.
- $\mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'] = \sigma^2\mathbf{I}_n$ becuase $\mathrm{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ as mentioned in (5) and $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$ as in (6), therefore, $\mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}] = \mathrm{E}[(\boldsymbol{\varepsilon} - \mathbf{0})(\boldsymbol{\varepsilon} - \mathbf{0})'] = \mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$.

The trace of $\mathbf{I} - \mathbf{H}$

$$\begin{aligned}\mathrm{tr}(\mathbf{H}) &= \mathrm{tr}\left(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\right) = \mathrm{tr}\left((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\right) \\ &= \mathrm{tr}(\mathbf{I}_p) = p, \\ \mathrm{tr}(\mathbf{I}_n - \mathbf{H}) &= n - p.\end{aligned} \tag{31}$$

$\mathrm{E}[\mathrm{SS}_{\mathrm{Res}}]$ and an unbiased estimator of σ^2

Applying (30) with $\mathbf{B} = \mathbf{I} - \mathbf{H}$ to (29), and using (31):

$$\mathrm{E}[\mathrm{SS}_{\mathrm{Res}}] = \sigma^2(n-p), \tag{32}$$

$$\hat{\sigma}^2 = \mathrm{MS}_{\mathrm{Res}} = \frac{\mathrm{SS}_{\mathrm{Res}}}{n-p}. \tag{33}$$

$\mathrm{E}[\mathrm{MS}_{\mathrm{Res}}] = \sigma^2$. For $p = 2$ this is the SLR estimator (2).

Question: why divide by $n - p$?

The n residuals are $e_i = y_i - \hat{y}_i$. Looking at (32), why is the divisor of SSE $n - p$ and not n? What is the bias of $\hat{\sigma}^2_{\mathrm{ML}}$ in (27)?

- The residuals satisfy the p linear constraints $\mathbf{X'e} = \mathbf{0}$ (as seen in euqation 19), so only $n - p$ of them are free, therefore, $n - p$ degrees of freedom
- $\mathrm{E}[\mathrm{SS}_{\mathrm{Res}}/n] = \sigma^2(n-p)/n$ which is smaller than σ^2
- So the ML estimator of the variance term, $\hat{\sigma}^2_{\mathrm{ML}} = \mathrm{SS}_{\mathrm{Res}}/n$ underestimates σ^2

Standard errors of the coefficients

Replace σ^2 by $\hat{\sigma}^2$ in (23) and (24):

$$\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2(\mathbf{X'X})^{-1},$$

$$\mathrm{se}(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 C_{jj}}. \tag{34}$$

Delivery time data: $\hat{\sigma}^2$

- $\mathrm{SS}_{\mathrm{Res}} = 233.73$, with $n - p = 25 - 3 = 22$ degrees of freedom.
- $\hat{\sigma}^2 = \mathrm{MS}_{\mathrm{Res}} = 233.73/22 = 10.62$.
- $\hat{\sigma} = 3.26$ minutes: the typical deviation of a delivery time from the fitted plane.

Delivery time data: variance-covariance matrix of the estimates

From (23) with σ^2 replaced by $\hat{\sigma}^2 = 10.62$ and $(\mathbf{X'X})^{-1}$ from before:

$$\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2(\mathbf{X'X})^{-1} = 10.62 \begin{pmatrix} 1.13 \times 10^{-1} & -4.45 \times 10^{-3} & -8.37 \times 10^{-5} \\ -4.45 \times 10^{-3} & 2.74 \times 10^{-3} & -4.79 \times 10^{-5} \\ -8.37 \times 10^{-5} & -4.79 \times 10^{-5} & 1.23 \times 10^{-6} \end{pmatrix}$$

$$\approx \begin{pmatrix} 1.20 & -4.73 \times 10^{-2} & -8.89 \times 10^{-4} \\ -4.73 \times 10^{-2} & 2.91 \times 10^{-2} & -5.09 \times 10^{-4} \\ -8.89 \times 10^{-4} & -5.09 \times 10^{-4} & 1.31 \times 10^{-5} \end{pmatrix}.$$

- The diagonal entries are $\widehat{\mathrm{Var}}(\hat{\beta}_j)$; their square roots are the standard errors 1.097, 0.171 and 0.00362 in (34).
- The off-diagonal entries are estimated covariances; e.g. $\widehat{\mathrm{Cov}}(\hat{\beta}_1, \hat{\beta}_2) = -5.09 \times 10^{-4} < 0$: the estimates of the cases and distance coefficients are negatively correlated, because the regressors are positively correlated ($r = 0.82$).

Delivery time data: standard errors

Using (34):

Coefficient	Estimate	Standard error
β_0 (intercept)	2.3412	1.0967
β_1 (cases)	1.6159	0.1707
β_2 (distance)	0.01438	0.003613

10 Coefficient of determination and adjusted R^2

Sums of squares

$$\begin{aligned} \mathrm{SS_T} &= \sum_{i=1}^{n}(y_i - \bar{y})^2 = \mathbf{y}'\mathbf{y} - n\bar{y}^2, \\ \mathrm{SS_{Res}} &= \mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y}, \\ \mathrm{SS_R} &= \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} - n\bar{y}^2, \\ \mathrm{SS_T} &= \mathrm{SS_R} + \mathrm{SS_{Res}}. \end{aligned} \tag{35}$$

The decomposition follows from the orthogonality (19).

The coefficient of determination

$$R^2 = \frac{\mathrm{SS_R}}{\mathrm{SS_T}} = 1 - \frac{\mathrm{SS_{Res}}}{\mathrm{SS_T}}. \tag{36}$$

- The proportion of the variability in y explained by the regressors; $0 \le R^2 \le 1$.
- Delivery time data: $R^2 = 0.9596$.

Adding an irrelevant regressor

What happens to R^2 when a regressor is added to the model? Does it depend on whether the regressor is relevant?

Add z, 25 independent $N(0,1)$ draws unrelated to the delivery time.

Model	p	R^2	MSE
cases	2	0.9305	17.484
cases, distance	3	0.9596	10.624
cases, distance, z	4	0.9597	11.109

R^2 increases slightly with the useless regressor, while the MSE goes up.

Adjusted R^2

$$R^2_{\text{adj}} = 1 - \frac{\text{SS}_{\text{Res}}/(n-p)}{\text{SS}_{\text{T}}/(n-1)} = 1 - \frac{\text{MS}_{\text{Res}}}{\text{MS}_{\text{T}}}, \tag{37}$$

$$= 1 - \frac{n-1}{n-p}(1-R^2). \tag{38}$$

$\text{MS}_{\text{T}} = \text{SS}_{\text{T}}/(n-1)$ is the sample variance of y.

Why adjusted R^2 penalises useless regressors

- MS_{T} does not depend on the model, so by (37) R^2_{adj} increases if and only if MS_{Res} decreases.
- Adding a regressor lowers SSE but also lowers the degrees of freedom $n-p$ in (33).
- The MSE decreases only if the drop in SSE outweighs the lost degree of freedom.

Adjusted R^2 in the example

Model	p	R^2	R^2_{adj}
cases	2	0.9305	0.9275
cases, distance	3	0.9596	0.9559
cases, distance, z	4	0.9597	0.9539

R^2_{adj} rises when the relevant regressor (distance) is added and falls when the irrelevant z is added.

Properties of adjusted R^2

From (38):

- For $p > 1$, $R^2_{\text{adj}} < R^2$ unless $R^2 = 1$.
- It can decrease when a regressor is added.
- It can be negative: $R^2_{\text{adj}} < 0$ whenever $R^2 < (p-1)/(n-1)$.

However,

- A high R^2 or R^2_{adj} does not mean the model is adequate: the assumptions must be checked with residual analysis.
- R^2_{adj} is a descriptive measure, not a hypothesis test.
- Formal comparison of models is done later with F-tests and information criteria.

11 SLR as a Special Case of MLR

SLR as a special Case of MLR

For SLR, $\mathbf{X} = [\mathbf{1}\ \ \mathbf{x}]$ and

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}, \qquad \mathbf{X}'\mathbf{y} = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix}.$$

Since $n\sum x_i^2 - (\sum x_i)^2 = nS_{xx}$,

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{nS_{xx}} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix}.$$

The second entry of $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ is

$$\hat{\beta}_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{nS_{xx}} = \frac{nS_{xy}}{nS_{xx}} = \frac{S_{xy}}{S_{xx}},$$

which is (1).

SLR variance from the MLR covariance matrix

In equation (23) we found that $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$.

Using the inverse computed for SLR, with $\sum x_i^2 = S_{xx} + n\bar{x}^2$:

$$\begin{aligned} \mathrm{Var}(\hat{\beta}_1) &= \sigma^2 \frac{n}{nS_{xx}} = \frac{\sigma^2}{S_{xx}}, \\ \mathrm{Var}(\hat{\beta}_0) &= \sigma^2 \frac{\sum x_i^2}{nS_{xx}} = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right), \\ \mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) &= -\sigma^2 \frac{\sum x_i}{nS_{xx}} = -\sigma^2 \frac{\bar{x}}{S_{xx}}. \end{aligned}$$

These are the familiar SLR results.

12 Summary

Summary

- Model: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, with $\mathrm{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$ (4).
- OLS: $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ (10); $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ is a projection onto $\mathrm{col}(\mathbf{X})$.
- $\hat{\boldsymbol{\beta}}$ is unbiased with $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ (23), and is BLUE.
- $\hat{\sigma}^2 = \mathrm{SS}_{\mathrm{Res}}/(n-p)$ is unbiased for σ^2 (33).
- R^2 never decreases when regressors are added; R^2_{adj} (37) penalises the loss of degrees of freedom.

Next lecture

- Inference on the regression coefficients: t-tests and confidence intervals.
- Test for significance of regression (F-test).
- Reading: Montgomery, Peck & Vining, Chapter 3.

A Proof of the Gauss–Markov Theorem

Gauss–Markov: setup of the proof

Any linear estimator can be written

$$\tilde{\boldsymbol{\beta}} = [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}]\mathbf{y}$$

for some $p \times n$ matrix $\mathbf{D}$. Its expectation is $\boldsymbol{\beta} + \mathbf{D}\mathbf{X}\boldsymbol{\beta}$, so unbiasedness for every $\boldsymbol{\beta}$ requires

$$\mathbf{D}\mathbf{X} = \mathbf{0}. \tag{39}$$

Gauss–Markov: the variance comparison

$$\begin{aligned}
\operatorname{Var}(\tilde{\boldsymbol{\beta}}) &= \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}][\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{D}'] \\
&= \sigma^2[(\mathbf{X}'\mathbf{X})^{-1} + (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{D}\mathbf{X})' \\
&\qquad + \mathbf{D}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{D}\mathbf{D}'] \\
&= \sigma^2(\mathbf{X}'\mathbf{X})^{-1} + \sigma^2\mathbf{D}\mathbf{D}',
\end{aligned}$$

where the last step uses (39).

Gauss–Markov: conclusion

- $\mathbf{a}'\mathbf{D}\mathbf{D}'\mathbf{a} = \|\mathbf{D}'\mathbf{a}\|^2 \geq 0$ for every $\mathbf{a}$, so $\mathbf{D}\mathbf{D}'$ is positive semidefinite.
- Hence $\operatorname{Var}(\tilde{\boldsymbol{\beta}}) - \operatorname{Var}(\hat{\boldsymbol{\beta}})$ is positive semidefinite, and $\operatorname{Var}(\mathbf{c}'\tilde{\boldsymbol{\beta}}) \geq \operatorname{Var}(\mathbf{c}'\hat{\boldsymbol{\beta}})$ for every $\mathbf{c}$.
- Equality holds only for $\mathbf{D} = \mathbf{0}$, i.e. $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}$.